

NIMEDINE Imipenem + Cilastatin (500+500) mg/vial Powder for Solution for Infusion

Imipenem and Cilastatin (500+500) mg

Product Description

White to almost white or light (pale) yellow powder.

After dilution: Range from colorless to yellow. Variations of color within this range do not affect the potency of the product.

Composition

Each vial contains 530 mg of Imipenem monohydrate and 530 mg of Cilastatin sodium corresponding to 500 mg of Imipenem and 500 mg of Cilastatin.

Pharmacodynamics

Pharmacotherapeutic group: Antibacterials for systemic use, carbapenems, ATC code: J01D H51

Mechanism of action

NIMEDINE consists of two components: imipenem and cilastatin sodium in a 1:1 ratio by weight.

Imipenem, also referred to as N-formimidoyl-thienamycin, is a semi-synthetic derivative of thienamycin, the parent compound produced by the filamentous bacterium *Streptomyces cattleya*.

Imipenem exerts its bactericidal activity by inhibiting bacterial cell wall synthesis in Gram-positive and Gram-negative bacteria through binding to penicillin-binding proteins (PBPs).

Cilastatin sodium is a competitive, reversible and specific inhibitor of dehydropeptidase-I, the renal enzyme which metabolizes and inactivates imipenem. It is devoid of intrinsic antibacterial activity and does not affect the antibacterial activity of imipenem.

Pharmacokinetic/Pharmacodynamic (PK/PD) relationship

Similar to other beta-lactam antibacterial agents, the time that imipenem concentrations exceed the MIC (T>MIC) has been shown to best correlate with efficacy.

Mechanism of resistance

Resistance to imipenem may be due to the following:

- Decreased permeability of the outer membrane of Gram-negative bacteria (due to diminished production of porins)
- Imipenem may be actively removed from the cell with an efflux pump
- Reduced affinity of PBPs to imipenem
- Imipenem is stable to hydrolysis by most beta-lactamases, including penicillinases and cephalosporinases produced by gram-positive and gram-negative bacteria, with the exception of relatively rare carbapenem hydrolysing beta-lactamases. Species resistant to other carbapenems do generally express co-resistance to imipenem. There is no target-based cross-resistance between imipenem and agents of the quinolone, aminoglycoside, macrolide and tetracycline classes

Breakpoints

EUCAST MIC breakpoints for imipenem to separate susceptible (S) pathogens from resistant (R) pathogens are as follows (v 1,1 2010-04-27):

- *Enterobacteriaceae* ¹: S ≤2 mg/l, R >8 mg/l
- *Pseudomonas* spp. ²: S ≤4 mg/l, R >8 mg/l
- *Acinetobacter* spp.: S ≤2 mg/l, R >8 mg/l
- *Staphylococcus* spp. ³: Inferred from ceftazidime susceptibility
- *Enterococcus* spp.: S ≤4 mg/l, R >8 mg/l
- *Streptococcus* A, B, C, G: The beta-lactam susceptibility of beta-haemolytic *streptococcus* groups A, B, C and G is inferred from the penicillin susceptibility
- *Streptococcus pneumoniae* ⁴: S ≤2 mg/l, R >2 mg/l
- Other streptococci ⁴: S ≤2 mg/l, R >2 mg/l
- *Haemophilus influenzae* ⁴: S ≤2 mg/l, R >2 mg/l
- *Moraxella catarrhalis* ⁴: S ≤2 mg/l, R >2 mg/l
- *Neisseria gonorrhoeae*: There is insufficient evidence that *Neisseria gonorrhoeae* is a good target for therapy with imipenem
- Gram-positive anaerobes: S ≤2 mg/l, R >8 mg/l
- Gram-negative anaerobes: S ≤2 mg/l, R >8 mg/l
- Non-species related breakpoints ⁵: S ≤2 mg/l, R >8 mg/l

¹ *Proteus* and *Morganella* species are considered poor targets for imipenem.

² The breakpoints for *Pseudomonas* relate to high dose frequent therapy (1g every 6 hours).

³ Susceptibility of staphylococci to carbapenems is inferred from the ceftazidime susceptibility.

⁴ Strains with MIC values above the susceptible breakpoint are very rare or not yet reported. The identification and antimicrobial susceptibility tests on any such isolate must be repeated and if the result is confirmed the isolate must be sent to a reference laboratory. Until there is evidence regarding clinical response for confirmed isolates with MIC above the current resistant breakpoint they should be reported resistant.

⁵ Non-species related breakpoint have been determined mainly on the basis of PK/PD data and are independent of MIC distributions of specific species. They are for use only for species not mentioned in the overview of species-related breakpoints or footnotes.

Susceptibility

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Commonly susceptible species:
Gram-positive aerobes: <i>Enterococcus faecalis</i> <i>Staphylococcus aureus</i> (Methicillin-susceptible)* <i>Staphylococcus coagulans</i> negative (Methicillin-susceptible) <i>Streptococcus agalactiae</i> <i>Streptococcus pneumoniae</i> <i>Streptococcus pyogenes</i> <i>Streptococcus viridans</i> group

Gram-negative aerobes: <i>Citrobacter freundii</i> <i>Enterobacter aerogenes</i> <i>Enterobacter cloacae</i> <i>Escherichia coli</i> <i>Haemophilus influenzae</i> <i>Klebsiella oxytoca</i> <i>Klebsiella pneumoniae</i> <i>Moraxella catarrhalis</i> <i>Serratia marcescens</i>
Gram-positive anaerobes: <i>Clostridium perfringens</i> ** <i>Peptostreptococcus</i> spp.**
Gram-negative anaerobes: <i>Bacteroides fragilis</i> <i>Bacteroides fragilis</i> group <i>Fusobacterium</i> spp. <i>Porphyromonas asaccharolytica</i> <i>Prevotella</i> spp. <i>Veillonella</i> spp.
Species for which acquired resistance may be a problem:
Gram-negative aerobes: <i>Acinetobacter baumannii</i> <i>Pseudomonas aeruginosa</i>
Inherently resistant species:
Gram positive aerobes: <i>Enterococcus faecium</i>
Gram-negative aerobes: Some strains of <i>Burkholderia cepacia</i> (formerly <i>Pseudomonas cepacia</i>) <i>Legionella</i> spp. <i>Stenotrophomonas maltophilia</i> (formerly <i>Xanthomonas maltophilia</i> , formerly <i>Pseudomonas maltophilia</i>)
Others:
<i>Chlamydia</i> spp. <i>Chlamydophila</i> spp. <i>Mycoplasma</i> spp. <i>Ureoplasma urealyticum</i>

* All methicillin-resistant staphylococci are resistant to NIMEDINE.

** EUCAST non-species related breakpoint is used.

Pharmacokinetics

Imipenem

Plasma concentrations

In normal volunteers, intravenous infusion of NIMEDINE over 20 minutes resulted in peak plasma levels of imipenem ranging from 12 to 20 µg/ml for the 250 mg/250 mg dose, from 21 to 58 µg/ml for the 500 mg/500 mg dose, and from 41 to 83 µg/ml for the 1000 mg/1000 mg dose. The mean peak plasma levels of imipenem following the 250 mg/250 mg, 500 mg/500 mg, and 1000 mg /1000

mg doses were 17, 39, and 66 µg/ml, respectively. At these doses, plasma levels of imipenem decline to below 1 µg/ml or less in four to six hours.

Distribution

The binding of imipenem to human serum proteins is approximately 20%.

Biotransformation and elimination

When administered alone, imipenem is metabolized in the kidneys by dehydropeptidase-I. Individual urinary recoveries ranged from 5 to 40%, with an average recovery of 15-20% in several studies.

Cilastatin is a specific inhibitor of dehydropeptidase-I enzyme and effectively inhibits metabolism of imipenem so that concomitant administration of imipenem and cilastatin allows therapeutic antibacterial levels of imipenem to be attained in both urine and plasma.

The plasma half-life of imipenem was one hour. Approximately 70% of the administered antibiotic was recovered intact in the urine within ten hours, and no further urinary excretion of imipenem was detectable. Urine concentrations of imipenem exceeded 10 µg/ml for up to eight hours after a 500 mg/500 mg dose of NIMEDINE. The remainder of the administered dose was recovered in the urine as antibacterially inactive metabolites, and faecal elimination of imipenem was essentially nil.

No accumulation of imipenem in plasma or urine has been observed with regimens of NIMEDINE, administered as frequently as every six hours, in patients with normal renal function.

Cilastatin

Plasma concentrations

Peak plasma levels of cilastatin, following a 20 minute intravenous infusion of NIMEDINE, ranged from 21 to 26 µg/ml for the 250 mg/250 mg dose, from 21 to 55 µg/ml for the 500 mg/500 mg dose and from 56 to 88 µg/ml for the 1000 mg/1000 mg dose. The mean peak plasma levels of cilastatin following the 250 mg/250 mg, 500 mg/500 mg, and 1000 mg/1000 mg doses were 22, 42, and 72 µg/ml respectively.

Distribution

The binding of cilastatin to human serum proteins is approximately 40%.

Biotransformation and elimination

The plasma half-life of cilastatin is approximately one hour. Approximately 70-80% of the dose of cilastatin was recovered unchanged in the urine as cilastatin within 10 hours of administration of NIMEDINE. No further cilastatin appeared in the urine thereafter. Approximately 10% was found as the N-acetyl metabolite, which has inhibitory activity against dehydropeptidase comparable to that of cilastatin. Activity of dehydropeptidase-I in the kidney returned to normal levels shortly after the elimination of cilastatin from the blood stream.

Renal insufficiency

Following a single 250 mg/250 mg intravenous dose of NIMEDINE, the area under the curve (AUCs) for imipenem increased 1.1-fold, 1.9-fold, and 2.7-fold in subjects with mild (Creatinine Clearance (CrCL) 50-80 ml/min/1.73 m²), moderate (CrCL 30-<50 ml/min/1.73 m²), and severe (CrCL <30 ml/min/1.73 m²) renal impairment, respectively, compared to subjects with normal renal function (CrCL >80 ml/min/1.73 m²), and AUCs for cilastatin increased 1.6-fold, 2.0-fold, and 6.2-fold in

subjects with mild, moderate, and severe renal impairment, respectively, compared to subjects with normal renal function. Following a single 250 mg/250 mg intravenous dose of NIMEDINE given 24 hours after haemodialysis, AUCs for imipenem and cilastatin were 3.7-fold and 16.4-fold higher, respectively, as compared to subjects with normal renal function. Urinary recovery, renal clearance and plasma clearance of imipenem and cilastatin decrease with decreasing renal function following intravenous administration of NIMEDINE. Dose adjustment is necessary for patients with impaired renal function (see Recommended Dose).

Hepatic insufficiency

The pharmacokinetics of imipenem in patients with hepatic insufficiency have not been established. Due to the limited extent of hepatic metabolism of imipenem, its pharmacokinetics are not expected to be affected by hepatic impairment. Therefore, no dose adjustment is recommended in patients with hepatic impairment (see Recommended Dose).

Paediatric patients

The average clearance (CL) and volume of distribution (Vdss) for imipenem were approximately 45% higher in paediatric patients (3 months to 14 years) as compared to adults. The AUC for imipenem following administration of 15/15 mg/kg per body weight of NIMEDINE to paediatric patients was approximately 30% higher than the exposure in adults receiving a 500 mg/500 mg dose. At the higher dose, the exposure following administration of 25/25 mg/kg NIMEDINE to children was 9% higher as compared to the exposure in adults receiving a 1000 mg/1000 mg dose.

Elderly

In healthy elderly volunteers (65 to 75 years of age with normal renal function for their age), the pharmacokinetics of a single dose of NIMEDINE 500 mg/500 mg administered intravenously over 20 minutes were consistent with those expected in subjects with slight renal impairment for which no dose alteration is considered necessary. The mean plasma half-lives of imipenem and cilastatin were 91 ± 7.0 minutes and 69 ± 15 minutes, respectively. Multiple dosing has no effect on the pharmacokinetics of either imipenem or cilastatin, and no accumulation of NIMEDINE was observed (see Recommended Dose).

Indication

TREATMENT

The activity of Nimedine against an unusually broad spectrum of pathogens makes it particularly useful in the treatment of polymicrobial and mixed aerobic/anaerobic infections, as well as initial therapy prior to the identification of the causative organisms. Nimedine is indicated for the treatment of the following infections due to susceptible organisms:

- Intra-abdominal infections
- Lower respiratory tract infections
- Gynecological infections
- Septicemia
- Genitourinary tract infections
- Bone and joint infections

- Skin and soft tissue infections
- Endocarditis

Nimedine is indicated for the treatment of mixed infections caused by susceptible strains of aerobic and anaerobic bacteria. The majority of these mixed infections are associated with contamination by fecal flora or flora originating from the vagina, skin and mouth. In these mixed infections, *Bacteroides fragilis* is the most commonly encountered anaerobic pathogen and is usually resistant to aminoglycosides, cephalosporins and penicillins. However, *Bacteroides fragilis* is usually susceptible to Nimedine.

Nimedine has demonstrated efficacy against many infections caused by aerobic and anaerobic gram-positive and gram-negative bacteria resistant to the cephalosporins, including cefazolin, cefoperazone, cephalothin, ceftazidime, cefotaxime, moxalactam, cefamandole, ceftazidime and ceftriaxone. Similarly, many infections caused by organisms resistant to aminoglycosides (gentamicin, amikacin, tobramycin) and/or penicillins (ampicillin, carbenicillin, penicillin-G, ticarcillin, piperacillin, azlocillin, mezlocillin) responded to treatment with Nimedine.

Nimedine is not indicated for the treatment of meningitis.

Recommended Dose

Nimedine is available as intravenous infusion only.

The dosage recommendations for Nimedine represent the quantity of imipenem to be administered. An equivalent amount of cilastatin is also present.

The total daily dosage of Nimedine should be based on the type or severity of infection and given in equally divided doses based on consideration of degree of susceptibility of the pathogen(s) and renal function.

Dosage in Adults

For Intravenous Injection Only

- The dosage of Nimedine in adult patients should be based on suspected or confirmed pathogen susceptibility as shown in Table 1 below. The dosage recommendations for Nimedine represent the quantity of imipenem to be administered. An equivalent amount of cilastatin is also present in the solution.
- These doses should be used for patients with creatinine clearance of ≥ 90 mL/min. A reduction in dose must be made for patients with creatinine clearance < 90 mL/min as shown in Table 2 (see Dosage in Adult Patients with Renal Impairment).
- Recommend that the maximum total daily dosage not exceed 4 g/day.
- Administer 500 mg by intravenous infusion over 20 to 30 minutes.
- Administer 1000 mg by intravenous infusion over 40 to 60 minutes.
- In patients who develop nausea during the infusion, the rate of infusion may be slowed.

Table 1: Dosage of Nimedine in Adult Patients with Creatinine Clearance ³ 90 mL/min

Suspected or Proven Pathogen Susceptibility	Dosage of Nimedine
If the infection is suspected or proven to be due to a susceptible bacterial species	500 mg every 6 hours OR 1000 mg every 8 hours
If the infection is suspected or proven to be due to bacterial species with intermediate susceptibility (see MICROBIOLOGY)	1000 mg every 6 hours

Dosage in Adult Patients with Renal Impairment

Patients with creatinine clearance less than 90 mL/min require dosage reduction of Nimedine as indicated in Table 2. The serum creatinine should represent a steady state of renal function. Use the Cockcroft-Gault method described below to calculate the creatinine clearance:

Males:

$$\frac{(\text{weight in kg}) \times (140 - \text{age in years})}{(72) \times \text{serum creatinine (mg/100 mL)}}$$

Females:

$$(0.85) \times (\text{value calculated for males})$$

Table 2: Dosage of Nimedine for Adult Patients in Various Renal Function Groups Based on Estimated Creatinine Clearance (CLcr)

	Creatinine clearance (mL/min)			
	≥ 90	60 – 90	30 – 60	15 – 30
Dosage of Nimedine *,† If the infection is suspected or proven to be due to a susceptible bacterial species:	500 mg every 6 hours	400 mg every 6 hours	300 mg every 6 hours	200 mg every 6 hours
	OR			
	1000 mg every 8 hours	500 mg every 6 hours	500 mg every 8 hours	500 mg every 12 hours
Dosage of Nimedine *,† If the infection is suspected or proven to be due to bacterial species with intermediate susceptibility (see MICROBIOLOGY):	1000 mg every 6 hours	750 mg every 8 hours	500 mg every 6 hours	500 mg every 12 hours

*Administer doses ≤ 500 mg by intravenous infusion over 20 to 30 minutes.

†Administer doses > 500 mg by intravenous infusion over 40 to 60 minutes.

In patients who develop nausea during the infusion, the rate of infusion may be slowed.

In patients with creatinine clearances of 15 - 30 mL/min, there may be an increased risk of seizures (see Warnings and Precautions). Patients with creatinine clearance less than 15 mL/min should not

receive Nimedine unless haemodialysis is instituted within 48 hours. There is inadequate information to recommend usage of Nimedine for patients undergoing peritoneal dialysis.

Dosage in Hemodialysis Patients

When treating patients with creatinine clearances of less than 15 mL/min who are undergoing hemodialysis, use the dosage recommendations for patients with creatinine clearances of 15 – 30 mL/min in Table 2 above. Both imipenem and cilastatin are cleared from the circulation during hemodialysis. The patient should receive Nimedine after hemodialysis and at intervals timed from the end of that hemodialysis session. Dialysis patients, especially those with background CNS disease, should be carefully monitored; for patients on hemodialysis, Nimedine is recommended only when the benefit outweighs the potential risk of seizures. (see Dosage in Adult Patients with Renal Impairment).

Dosage in Paediatric Patients (3 months or older)

Nimedine is not recommended in paediatric patients with CNS infections because of the risk of seizures (see Warnings and Precautions).

Nimedine is not recommended in paediatric patients <30 kg with renal impairment, as no data are available (see Warnings and Precautions).

Based on studies in adults, the maximum total daily dose in pediatric patients should not exceed 4 g/day (see Dosage in Adults).

The recommended dosage for pediatric patients with non-CNS infections is shown in Table 3 below:

Table 3: Recommended Nimedine Dosage in Pediatric Patients for Non-CNS Infections

Age	Dose (mg/kg) *,†	Frequency (hours)
≥ 3 Months of Age		
	15-25 mg/kg	Every 6 hours
≤ 3 months of age (≥ 1,500 g body weight)		
4 weeks to 3 months of age	25 mg/kg	Every 6 hours
1 to 4 weeks of age	25 mg/kg	Every 8 hours
Less than 1 week of age	25 mg/kg	Every 12 hours

* Doses ≤ 500 mg should be given by intravenous infusion over 20 to 30 minutes

† Doses > 500 mg should be given by intravenous infusion over 40 to 60 minutes

Recommend that the maximum total daily dosage not exceed 4g/day

Preparation of Nimedine for IV Administration

Nimedine is supplied as a dry powder in a single-dose vial that must be reconstituted and further diluted using aseptic technique prior to IV infusion as outlined below.

- To prepare the infusion solution, contents of the vial must be reconstituted by adding approximately 10 mL of the appropriate diluent to the vial. List of appropriate diluents are as follows:
 - 0.9% Sodium Chloride Injection
 - 5% Glucose Injection
- The content of the vial is transferred to 100 ml of appropriate infusion solution.
- Add approximately 10 ml of the appropriate infusion solution to the vial. Shake well and transfer the resulting mixture to the infusion solution container.
- Repeat the above procedure with additional 10 ml of infusion solution to ensure complete transfer of vial contents to the infusion solution. Agitate the resulting mixture until clear.
- The concentration of the reconstituted solution following the above procedure is approximately 5mg/ml for both Imipenem and Cilastatin.
- Reconstituted solutions of Nimedine range from colorless to yellow. Variations of color within this range do not affect the potency of the product.
- For patients with renal insufficiency, a reduced dose of Nimedine will be administered according to the patient's CrCl, as determined from Table 4. Prepare 100 mL of infusion solution as directed above. Select the volume (mL) of the final infusion solution needed for the appropriate dose of Nimedine as shown in Table 4.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. Discard if discoloration or visible particles are observed.

Table 4: Preparation of Nimedine Doses

Creatinine Clearance (mL/min)	Dosage of Nimedine (imipenem/cilastatin (mg))	Volume (mL) of Solution to be Removed and Discarded from Preparation	Volume (mL) of Final Infusion Solution Needed for Dosage
≥ 90	500/500	N/A	100
60 – 90	400/400	20	80
30 – 60	300/300	40	60
15 – 30	200/200	60	40

Route of Administration

Intravenous infusion

Contraindications

- Hypersensitivity to the active substances or to any of the excipients
- Hypersensitivity to any other carbapenem antibacterial agent
- Severe hypersensitivity (e.g. anaphylactic reaction, severe skin reaction) to any other type of beta-lactam antibacterial agent (e.g. penicillins or cephalosporins)

Warnings and Precautions

General

The selection of NIMEDINE to treat an individual patient should take into account the appropriateness of using a carbapenem antibacterial agent based on factors such as severity of the infection, the prevalence of resistance to other suitable antibacterial agents and the risk of selecting for carbapenem-resistant bacteria.

Hypersensitivity

Serious and occasionally fatal hypersensitivity (including anaphylactoid and severe cutaneous adverse reactions) reactions have been reported in patients receiving therapy with beta-lactams. These reactions are more likely to occur in individuals with a history of sensitivity to multiple allergens. Before initiating therapy with NIMEDINE, careful inquiry should be made concerning previous hypersensitivity reactions to carbapenems, penicillins, cephalosporins, other beta-lactams and other allergens. If an allergic reaction to NIMEDINE occurs, discontinue the therapy immediately.

Serious anaphylactic reactions require immediate emergency treatment.

Hepatic

Hepatic function should be closely monitored during treatment with NIMEDINE due to the risk of hepatic toxicity (such as increase in transaminases, hepatic failure and fulminant hepatitis).

Use in patients with liver disease: patients with pre-existing liver disorders should have liver function monitored during treatment with NIMEDINE. There is no dose adjustment necessary.

Haematology

A positive direct or indirect Coombs test may develop during treatment with NIMEDINE.

Antibacterial spectrum

The antibacterial spectrum of NIMEDINE should be taken into account especially in life-threatening conditions before embarking on any empiric treatment. Furthermore, due to the limited susceptibility of specific pathogens associated with e.g. bacterial skin and soft-tissue infections, to NIMEDINE, caution should be exercised. The use of NIMEDINE is not suitable for treatment of these types of infections unless the pathogen is already documented and known to be susceptible or there is a very high suspicion that the most likely pathogen(s) would be suitable for treatment. Concomitant use of an appropriate anti-MRSA agent may be indicated when MRSA infections are suspected or proven to be involved in the approved indications. Concomitant use of an aminoglycoside may be indicated when *Pseudomonas aeruginosa* infections are suspected or proven to be involved in the approved indications.

Interaction with valproic acid

The concomitant use of NIMEDINE and valproic acid/sodium valproate is not recommended.

Clostridium difficile

Antibiotic-associated colitis and pseudomembranous colitis have been reported with NIMEDINE and with nearly all other anti-bacterial agents and may range from mild to life-threatening in severity. It is important to consider this diagnosis in patients who develop diarrhoea during or after the use of NIMEDINE. Discontinuation of therapy with NIMEDINE and the administration of specific treatment for *Clostridium difficile* should be considered. Medicinal products that inhibit peristalsis should not be given.

Meningitis

NIMEDINE is not recommended for the therapy of meningitis.

Renal impairment

Imipenem-cilastatin accumulates in patients with reduced kidney function. CNS adverse reactions may occur if the dose is not adjusted to the renal function.

Central nervous system

CNS adverse reactions such as myoclonic activity, confusional states, or seizures have been reported, especially when recommended doses based on renal function and body weight were exceeded. These experiences have been reported most commonly in patients with CNS disorders (e.g. brain lesions or history of seizures) and/or compromised renal function in whom accumulation of the administered entities could occur. Hence close adherence to recommended dose schedules is urged especially in these patients. Anticonvulsant therapy should be continued in patients with a known seizure disorder.

Special awareness should be made to neurological symptoms or convulsions in children with known risk factors for seizures, or on concomitant treatment with medicinal products lowering the seizures threshold.

If focal tremors, myoclonus, or seizures occur, patients should be evaluated neurologically and placed on anticonvulsant therapy if not already instituted. If CNS symptoms continue, the dose of NIMEDINE should be decreased or discontinued.

Patients with creatinine clearances of ≤ 5 ml/min/1.73 m² should not receive NIMEDINE unless haemodialysis is instituted within 48 hours. For patients on haemodialysis, NIMEDINE is recommended only when the benefit outweighs the potential risk of seizures.

Paediatric population

Clinical data are insufficient to recommend the use of NIMEDINE in children under 1 year of age or paediatric patients with impaired renal function (serum creatinine >2 mg/dl). See also above under Central nervous system.

NIMEDINE 500 mg/500 mg contains approximately 1.6 mEq (approximately 37.6 mg) of sodium which should be taken into consideration by patients on a controlled sodium diet.

Interactions with Other Medicaments

Generalized seizures have been reported in patients who received ganciclovir and NIMEDINE. These medicinal products should not be used concomitantly unless the potential benefit outweighs the risks.

Decreases in valproic acid levels that may fall below the therapeutic range have been reported when valproic acid was co-administered with carbapenem agents. The lowered valproic acid levels can lead to inadequate seizure control; therefore, concomitant use of imipenem and valproic acid/sodium valproate is not recommended and alternative antibacterial or anti-convulsant therapies should be considered.

Oral anti-coagulants

Simultaneous administration of antibiotics with warfarin may augment its anti-coagulant effects. There have been many reports of increases in the anti-coagulant effects of orally administered anti-coagulant agents, including warfarin in patients who are concomitantly receiving antibacterial agents. The risk may vary with the underlying infection, age and general status of the patient so that

the contribution of the antibiotic to the increase in INR (international normalised ratio) is difficult to assess. It is recommended that the INR should be monitored frequently during and shortly after co-administration of antibiotics with an oral anti-coagulant agent.

Concomitant administration of NIMEDINE and probenecid resulted in minimal increases in the plasma levels and plasma half-life of imipenem. The urinary recovery of active (non-metabolized) imipenem decreased to approximately 60% of the dose when NIMEDINE was administered with probenecid. Concomitant administration of NIMEDINE and probenecid doubled the plasma level and half-life of cilastatin, but had no effect on urine recovery of cilastatin.

Pregnancy and Lactation

Pregnancy

There are no adequate and well-controlled studies for the use of NIMEDINE in pregnant women.

The potential risk for humans is unknown.

NIMEDINE should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Breastfeeding

Imipenem and cilastatin are excreted into the mother's milk in small quantities. Little absorption of either compound occurs following oral administration. Therefore it is unlikely that the suckling infant will be exposed to significant quantities. If the use of NIMEDINE is deemed necessary, the benefit of breast feeding for the child should be weighed against the possible risk for the child.

Side Effects

All adverse reactions are listed under system organ class and frequency: Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), Very rare ($< 1/10,000$) and not known (cannot be estimated from the available data).

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System Organ Class	Frequency	Event
Infections and infestations	Rare	pseudomembranous colitis, candidiasis
	Very rare	gastro-enteritis
Blood and lymphatic system disorders	Common	eosinophilia
	Uncommon	pancytopenia, neutropenia, leucopenia, thrombocytopenia, thrombocytosis
	Rare	agranulocytosis
	Very rare	haemolytic anaemia, bone marrow depression
Immune system disorders	Rare	anaphylactic reactions
Psychiatric disorders	Uncommon	psychic disturbances including hallucinations and confusional states
Nervous system disorders	Uncommon	seizures, myoclonic activity, dizziness, somnolence
	Rare	encephalopathy, paraesthesia, focal tremor, taste perversion
	Very rare	aggravation of myasthenia gravis, headache

Ear and labyrinth disorders	Rare	hearing loss
	Very rare	vertigo, tinnitus
Cardiac disorders	Very rare	cyanosis, tachycardia, palpitations
Vascular disorders	Common	thrombophlebitis
	Uncommon	hypotension
	Very rare	flushing
Respiratory, thoracic and mediastinal disorders	Very rare	dyspnoea, hyperventilation, pharyngeal pain
Gastrointestinal disorders	Common	diarrhoea, vomiting, nausea. Medicinal product-related nausea and/or vomiting appear to occur more frequently in granulocytopenic patients than in non-granulocytopenic patients treated with NIMEDINE
	Rare	staining of teeth and/or tongue
	Very rare	haemorrhagic colitis, abdominal pain, heartburn, glossitis, tongue papilla hypertrophy, increased salivation
Hepatobiliary disorders	Rare	hepatic failure, hepatitis
	Very rare	fulminant hepatitis
Skin and subcutaneous tissue disorders	Common	rash (e.g. exanthematous)
	Uncommon	urticaria, pruritus
	Rare	toxic epidermal necrolysis, angioedema, Stevens-Johnson syndrome, erythema multiforme, exfoliative dermatitis
	Very rare	hyperhidrosis, skin texture changes
Musculoskeletal and connective tissue disorders	Very rare	polyarthralgia, thoracic spine pain
Renal and urinary disorders	Rare	acute renal failure, oliguria/anuria, polyuria, urine discoloration (harmless and should not be confused with haematuria). The role of NIMEDINE in changes in renal function is difficult to assess, since factors predisposing to pre-renal azotemia or to impaired renal function usually have been present
Reproductive system and breast disorders	Very rare	pruritus vulvae
General disorders and administration site conditions	Uncommon	fever, local pain and induration at the injection site, erythema at the injection site
	Very rare	chest discomfort, asthenia/weakness
Investigations	Common	Increases in serum transaminases, increases in serum alkaline phosphatase
	Uncommon	A positive direct Coombs' test, prolonged prothrombin time, decreased haemoglobin, increases in serum bilirubin, elevations in

		serum creatinine, elevations in blood urea nitrogen
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Paediatric population

Paediatric (≥3 months of age)

Studies in paediatric patients ≥3 months of age, the reported adverse reactions were consistent with those reported for adults.

Symptoms and Treatment of Overdose

Symptoms of overdose that can occur are consistent with the adverse reaction profile; these may include seizures, confusion, tremors, nausea, vomiting, hypotension, and bradycardia. No specific information is available on treatment of overdose with NIMEDINE. Imipenem-cilastatin sodium is haemodialyzable. However, usefulness of this procedure in the overdose setting is unknown.

Effects on Ability to Drive and Use Machine

No studies on the effects on the ability to drive and use machines have been performed. However, there are some side effects (such as hallucination, dizziness, somnolence, and vertigo) associated with this product that may affect some patients' ability to drive or operate machinery.

Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in section Instruction for Use.

The product is chemically incompatible with lactate and must not be reconstituted in solutions that contain it. However, it can be administered into an IV tubing, through which a lactate solution is to be infused.

The product must not be mixed or physically added to other antibiotics.

Instructions for Use

Each vial is for single use only.

Reconstitution:

Contents of each vial must be transferred to 100 ml of an appropriate infusion solution: 0.9% sodium chloride. In exceptional circumstances where 0.9% sodium chloride cannot be used for clinical reasons 5% glucose may be used instead.

A suggested procedure is to add approximately 10 ml of the appropriate infusion solution to the vial. Shake well and transfer the resulting mixture to the infusion solution container.

CAUTION: THE MIXTURE IS NOT FOR DIRECT INFUSION.

Repeat with an additional 10 ml of infusion solution to ensure complete transfer of vial contents to the infusion solution. The resulting mixture should be agitated until clear.

The concentration of the reconstituted solution following the above procedure is approximately 5 mg/ml for both imipenem and cilastatin.

Variations of colour, from colourless to yellow, do not affect the potency of the product.

Any unused product or waste material should be disposed of in accordance with local requirements.

Storage Conditions

Do not store above 30 °C.

After reconstitution and dilution:

The reconstituted solution is stable for 2 hours when stored at 25°C.

Shelf Life

36 months.

After reconstitution and dilution:

The reconstituted solution is stable for 2 hours when stored at 25 °C.

Dosage Forms and Packaging Available

Powder for Solution for Infusion- Pack of 1 vial containing 500 mg imipenem and 500mg cilastatin.

Registration Number

NIMEDINE Imipenem + Cilastatin (500+500) mg/vial Powder for Solution for Infusion-
MAL23046015AZ

Manufacturer

ANFARM HELLAS S.A.,
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Schimatari Viotias, 32009, GREECE

Product Registration Holder

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Date of Revision

21st February 2023